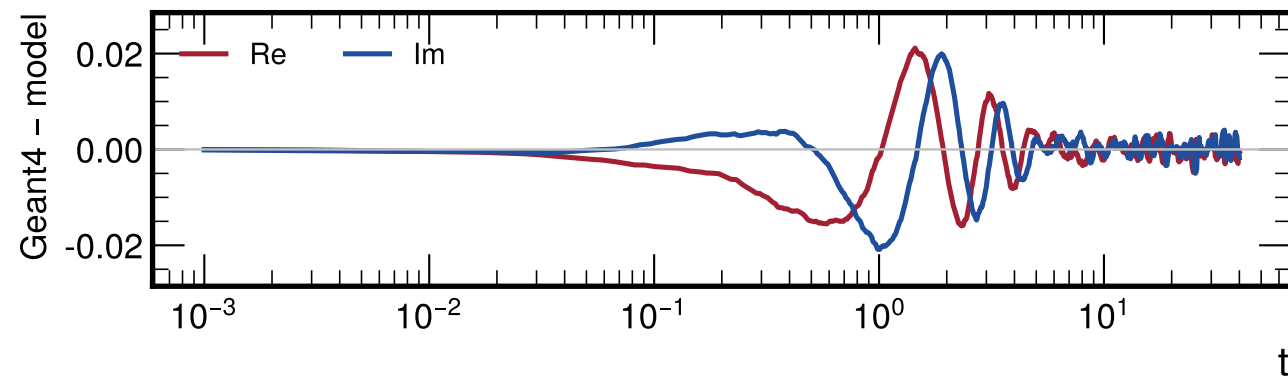
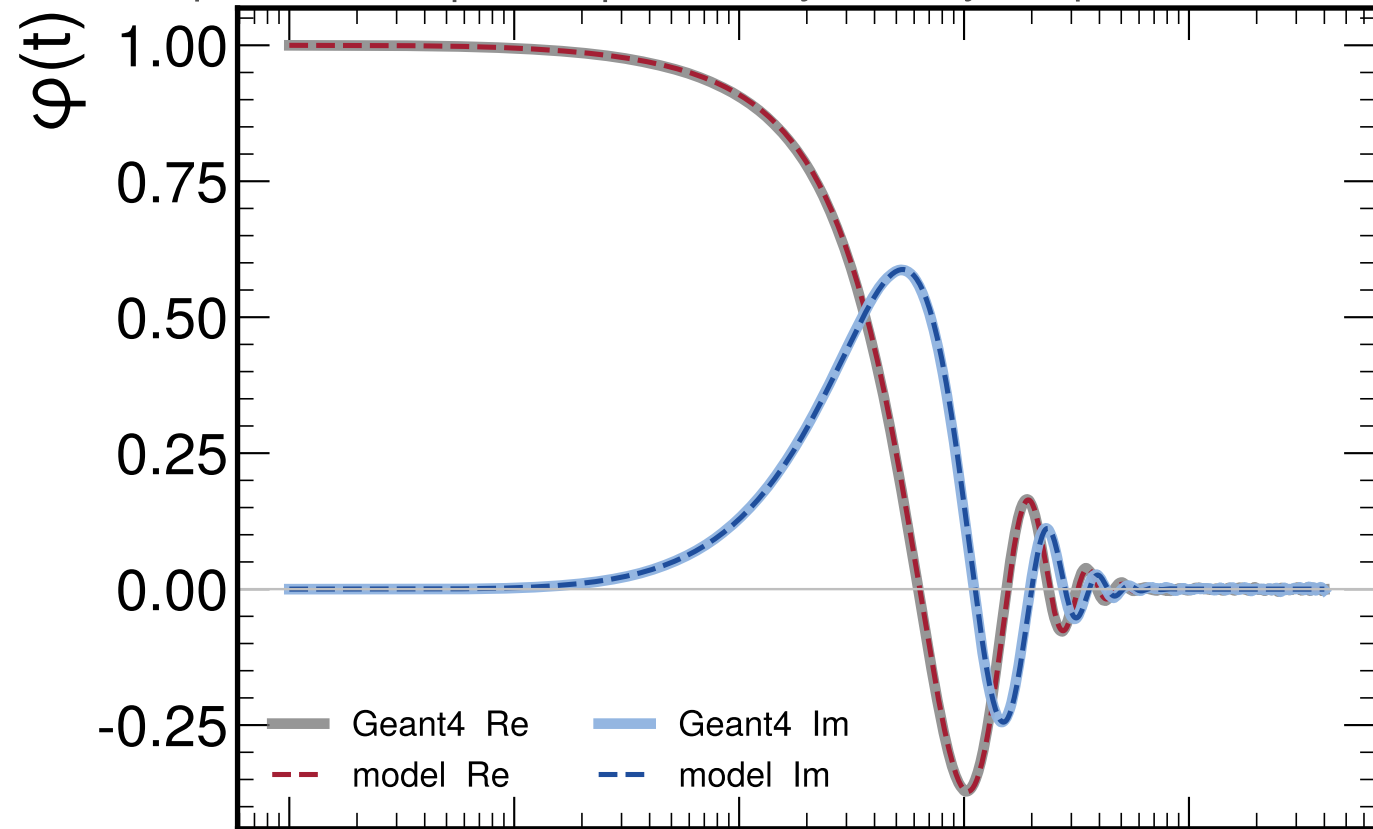
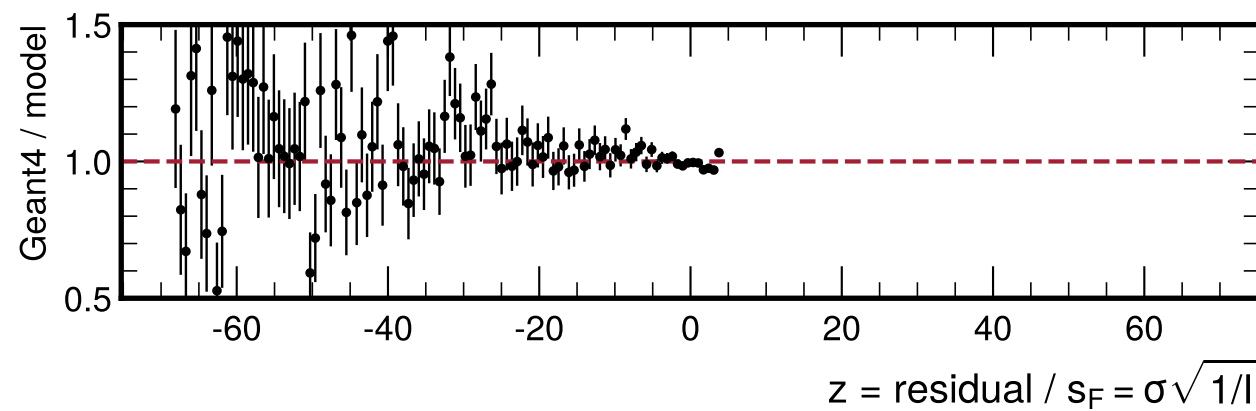
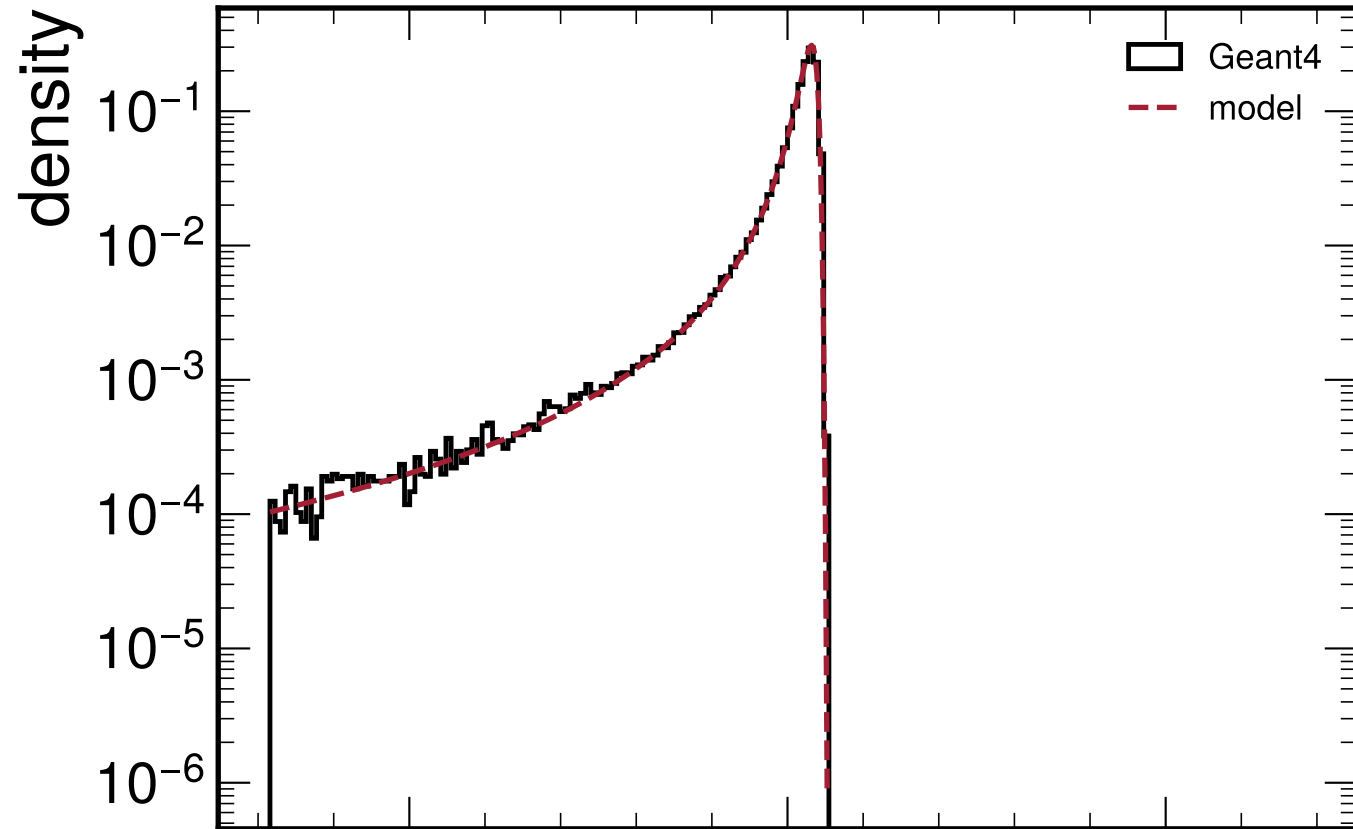


q/p at r = 107 cm — ionization straggling
 mu-, $p_T = 3$ GeV, $\eta = 0.3$, $\phi = 0.7$ · layered toy, 14 planes · radiation OFF



closure $\langle e^{-uz^2} \rangle_{\text{data}} - \langle e^{-uz^2} \rangle_{\text{model}}$
 $u = 1: -0.00011 \pm 0.00029$ (-0.4σ)
 peak $u = 0.03: -0.00164 \pm 0.00036$ rms 0.00100



$z = \text{residual} / s_F = \sigma \sqrt{1/l}$